Simulation Experiment 6

IPv4 and IPv6 Addressing, Subnetting and Connectivity Analysis using Cisco Packet Tracer

Objectives:

(i) To design and implement an IPv4 addressing scheme with subnetting (VLSM/fixed-length) and verify connectivity between subnets through a router.

(ii) To configure IPv6 global unicast addressing on a network and analyse end-to-end connectivity using ping, traceroute and routing-table verification.

Software/Apparatus required:

(i) Cisco Packet Tracer

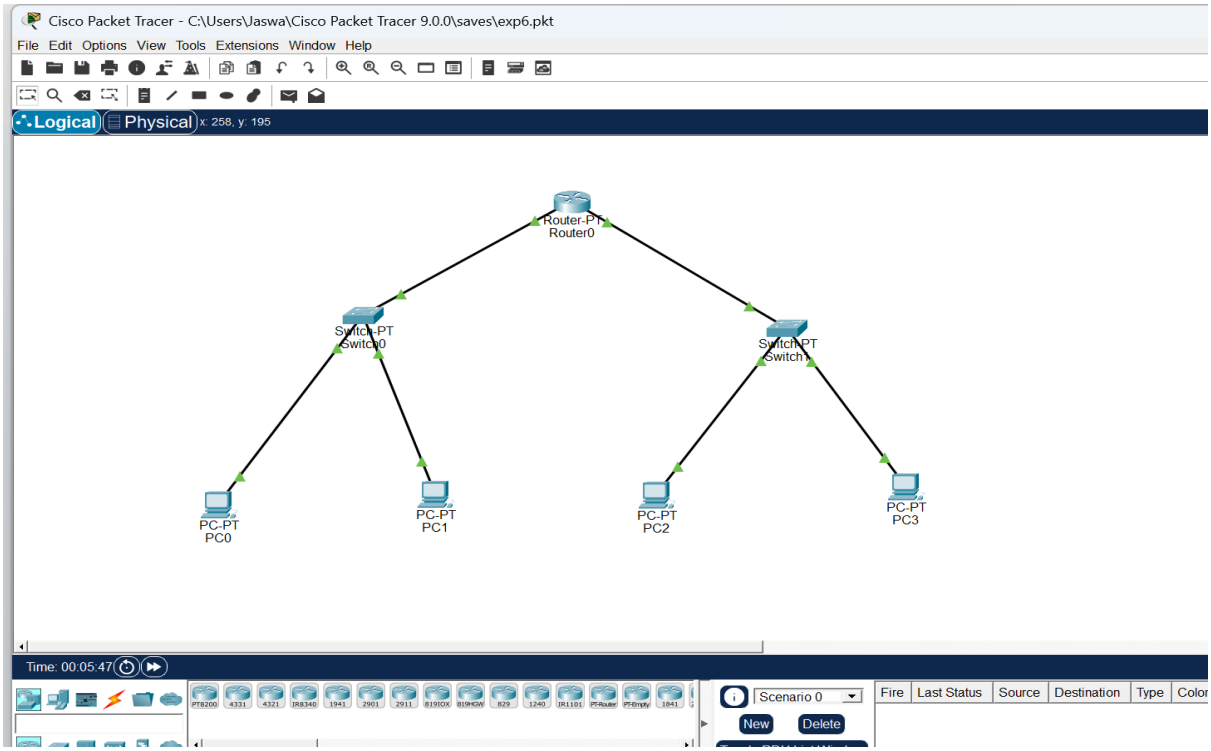
(ii) End Devices (PCs)

(iii)Switches (2960)

(iv)Router (2911 / ISR 4321)

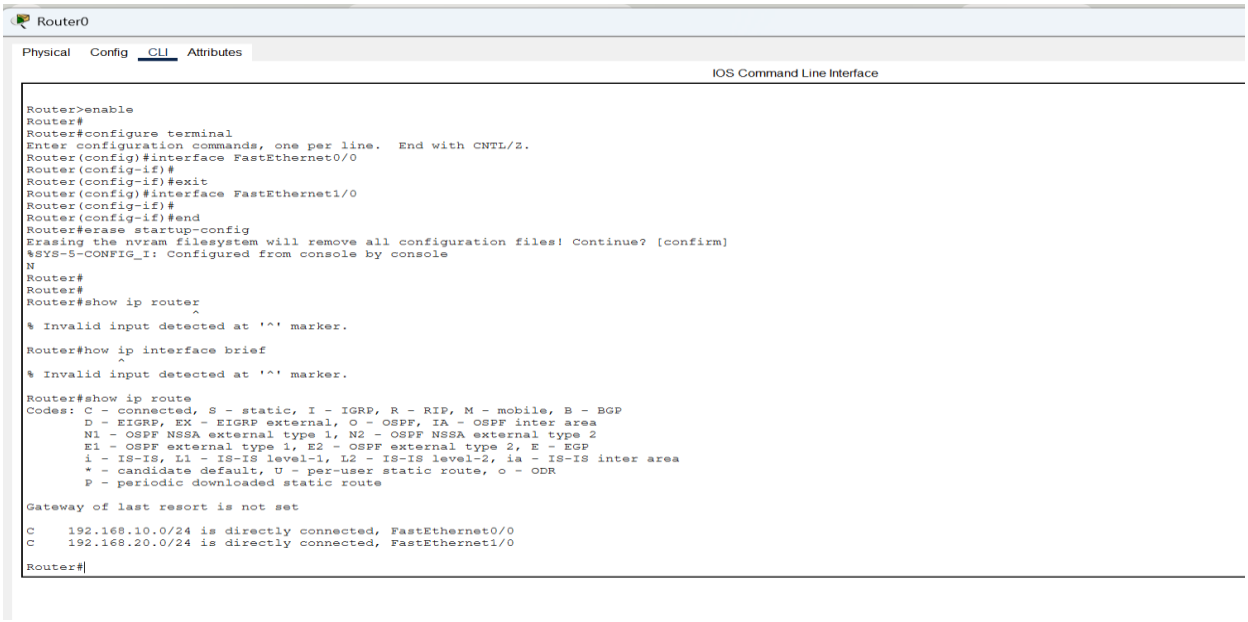
(v) Copper Straight-through Cable / Automatic Cable Selector

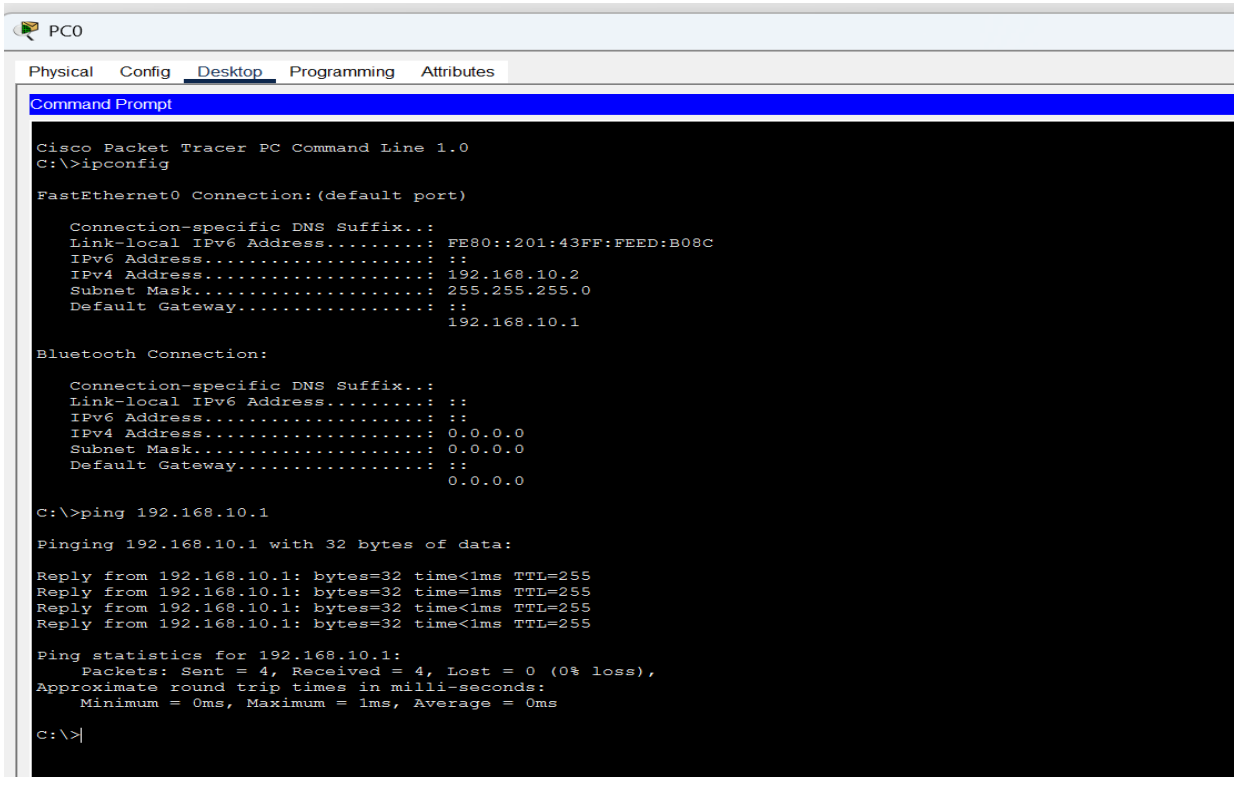
Network Diagram:



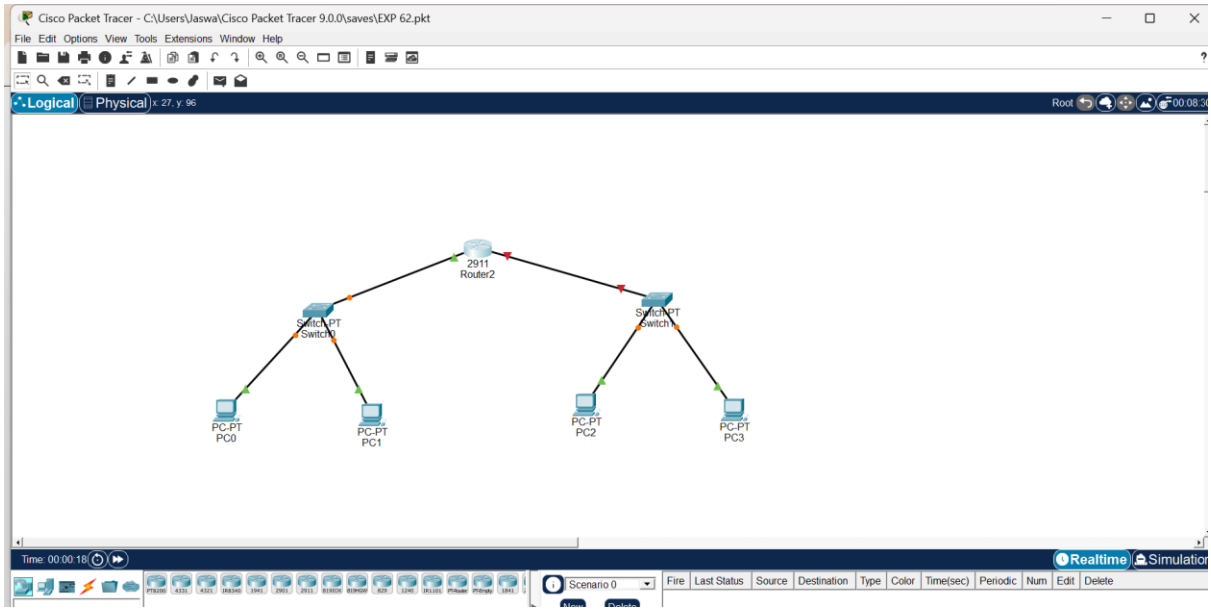
Device	IP address	Subnet mask	Default gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
PC2	192.168.20.2	255.255.255.0	192.168.20.1
PC3	192.168.20.3	255.255.255.0	192.168.20.1

Packet Tracer Command Prompt Output and Router Verification

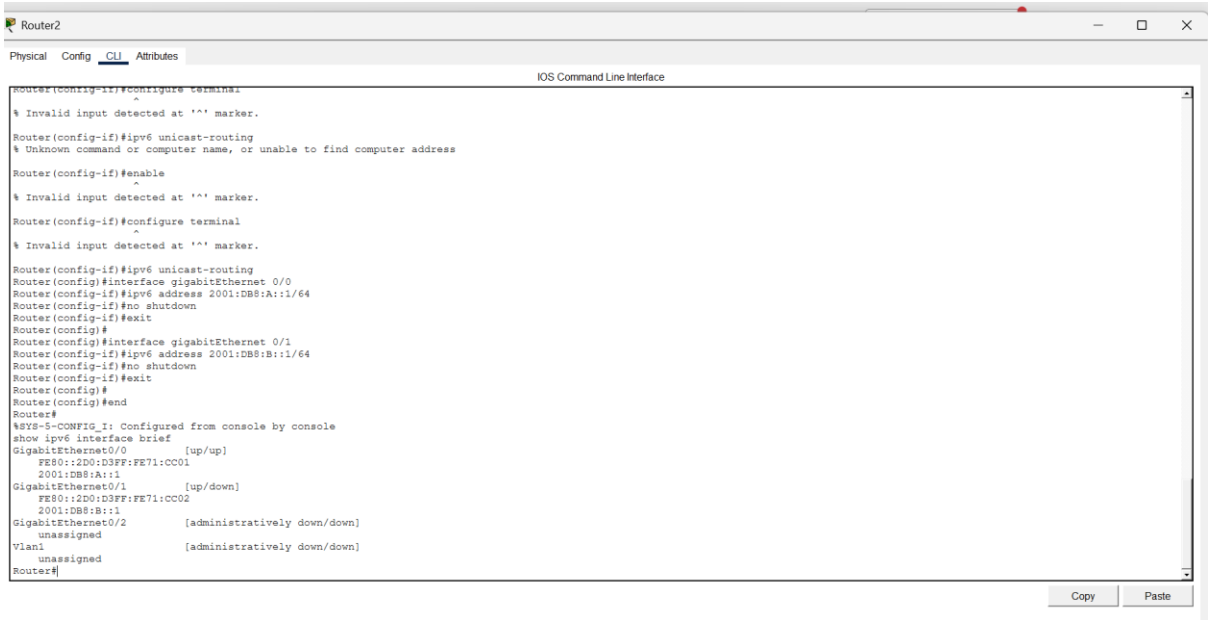


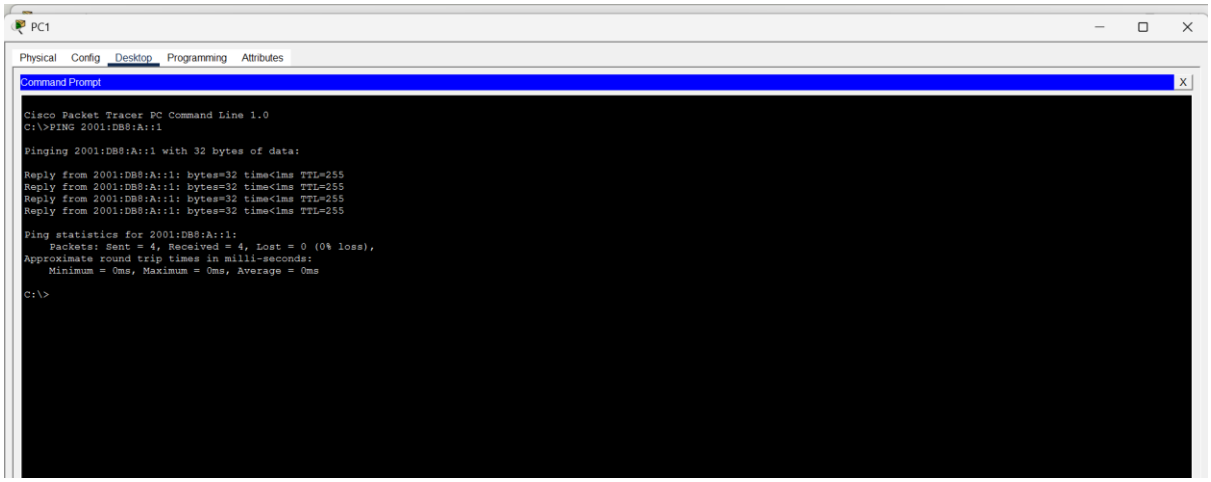


Objective 2: IPv6 Addressing



Device	IPv6 address	Prefix	Default gateway
PC4	2001:DB8:A::10	64	2001:DB8:A::1
PC5	2001:DB8:A::11	64	2001:DB8:A::1
PC6	2001:DB8:B::10	64	2001:DB8:B::1
PC7	2001:DB8:B::11	64	2001:DB8:B::1





Result:

Objective 1: IPv4 Subnetting

(i) The 192.168.10.0/24 network was successfully divided into /26 subnets, and devices in different subnets communicated through the router with 0% packet loss.

Objective 2: IPv6 Addressing

(ii) IPv6 global unicast addressing was configured and end-to-end connectivity between the two IPv6 networks was verified using ping and routing-table analysis.